

Lou Chang

Backend Engineer | APIs, Data Systems & Cloud Infrastructure

Toronto, ON, Canada

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Summary

Backend engineer with 5+ years building and stabilizing API and data systems across healthcare, finance, B2B, and legal workflows. Combines .NET/SQL depth with Python/FastAPI, Terraform, CI/CD, and AWS/Google Cloud delivery. Strongest where application behavior, database performance, and delivery infrastructure meet—cutting critical administration workflows from 5–8 minutes to 5 seconds (fastest 0.4s) and removing client-access bottlenecks for a 40–50 API financial platform.

Experience

Software Engineer — Backend & Cloud Infrastructure (Contract)

Independent Consulting

Toronto, ON
Sep. 2024 - Present

- Reworked a prototype into a maintainable FastAPI backend by separating domain, API, persistence, and provider boundaries, with SQLAlchemy/Alembic persistence, Shopify onboarding, and quality gates.
- Built dynamic template rendering for Ontario Form 13.1 by cloning and reindexing template rows, preserving copied-row formatting while supporting variable-length financial and property data without manual row creation; added behavior-focused transform tests.
- Built a Terraform deployment path on Google Cloud with Cloud Run, Cloud SQL, Secret Manager, GitHub Actions Workload Identity Federation, and Cloud Monitoring, replacing an EC2 host, NGINX, and Let's Encrypt with managed services while separating development and production resources.

Web Development Engineer

Duotify Inc.

Taipei, Taiwan
Jun. 2020 - Jan. 2024

- During annual conference registration season, cut accounting, event, and membership administration workflows from 5-8 minutes to 5 seconds, fastest 0.4 seconds, on a healthcare platform serving about 3,000 members by replacing multi-table LINQ embedded in AutoMapper with SQL Server procedures, views, and functions.
- Built the company's first AWS development and test environment for a 40-50 API financial platform, removing dependency on restricted client access and enabling local verification before delivering 4-5 APIs and completing on-site acceptance.
- Designed 20-30 REST APIs for a greenfield local-business marketing platform with JWT authentication and transaction-protected data consistency, establishing NSwag-documented contracts and Docker-based SQL Server development for frontend integration.
- Operated and improved bank-wide self-hosted GitLab CE serving about 100-150 engineers; built a production-like test environment to validate upgrades and automated log cleanup and routine maintenance with Bash and Cron.
- Mitigated a week-long traffic and scraper incident on a top-three Taiwan used-car marketplace with 50-70k monthly visits by using Azure Monitor to identify App Service saturation, configuring autoscaling, and supporting temporary traffic rerouting.

ERP Applications Engineer

Unitech Computer Co., Ltd.

Taipei, Taiwan
Mar. 2019 - May 2020

- Maintained product search, inventory, quotation, order, and member workflows for a nationwide B2B platform used by resellers, enterprise procurement teams, and internal sales across AngularJS, ASP.NET MVC, Oracle, SQL Server, and PostgreSQL.
- Resolved production defects, long-standing bugs, and data-query issues across the legacy web and multi-database stack, sustaining daily ordering operations under active business use.
- Built an Angular 8 and ASP.NET MVC 5 modernization PoC with NG-ZORRO and presented it to the ERP engineering department as a reference for updating the legacy AngularJS interface.

Education & Certification

Seneca Polytechnic

Diploma, Computer Programming

Toronto, Canada
May 2024 - Aug. 2025

Hwa Hsia University of Technology

Bachelor's, Information Management

New Taipei City, Taiwan
Sep. 2018 - Jun. 2020

Microsoft

AZ-900: Azure Fundamentals

Taipei, Taiwan
Feb. 2022

Technologies

Backend & Data APIs & Quality Cloud & Delivery

Python, FastAPI, C#, .NET, Microsoft SQL Server, PostgreSQL, Oracle Database, SQLAlchemy
REST APIs, OpenAPI / NSwag, JWT, pytest, Pyright, Ruff, Docker
AWS, Google Cloud, Cloud Run, Terraform, GitHub Actions, GitLab CE, Workload Identity Federation